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===== CASE 1 =====
Original Dataset
Available Water: 13,500 L

=====
ALLOCATION RESULT
=====
ID Zone Request Minimum Priority Allocated Delivered Shortage Status
-----
Z01 Municipal Hospital 4000 3000 40.00 3477 3408 592 MINIMUM
Z04 Ward 5 Residential Area 3800 1800 35.00 2045 1800 2000 MINIMUM
Z02 Central Flood Shelter 3500 2500 35.00 2604 2500 1000 MINIMUM
Z05 School Emergency Shelter 2600 1600 35.00 1684 1600 1000 MINIMUM
Z06 Fire and Emergency Service 2000 1500 35.00 1515 1500 500 MINIMUM
Z03 Ward 3 Residential Area 4500 2000 30.00 2174 2000 2500 MINIMUM
=====

Priority Order:
1. Z01 - Municipal Hospital (Priority 40.00)
2. Z04 - Ward 5 Residential Area (Priority 35.00)
3. Z02 - Central Flood Shelter (Priority 35.00)
4. Z05 - School Emergency Shelter (Priority 35.00)
5. Z06 - Fire and Emergency Service (Priority 35.00)
6. Z03 - Ward 3 Residential Area (Priority 30.00)

Summary
-----
Total Requested Water : 20400.00 L
Total Allocated Water : 13500.00 L
Total Effective Water Delivered: 12807.56 L
Total Unresolved Requirement : 7592.44 L
Remaining Water : 0.00 L

Service Conditions
-----
Full Requirement Satisfied : 0 zone(s)
Minimum Requirement Satisfied: 6 zone(s)
Below Minimum : 0 zone(s)
Completely Unserved : 0 zone(s)

Check: total allocated (13500.00 L) does not exceed available supply (13500.00 L). OK.

Zone with the highest unresolved requirement: Z03 (Ward 3 Residential Area) - shortage 2500.00 L

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EMERGENCY POTABLE WATER ALLOCATION
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===== CASE 2 =====
Sufficient Water - All Zones Receive Full Requested Quantity

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===== CASE 2 =====
Sufficient Water - All Zones Receive Full Requested Quantity
Available Water: 22,000 L

=====
ALLOCATION RESULT
=====
ID Zone Request Minimum Priority Allocated Delivered Shortage Status
-----
Z01 Municipal Hospital 4000 3000 40.00 4082 4000 0 FULL
Z04 Ward 5 Residential Area 3800 1800 35.00 4318 3800 0 FULL
Z02 Central Flood Shelter 3500 2500 35.00 3646 3500 0 FULL
Z05 School Emergency Shelter 2600 1600 35.00 2737 2600 0 FULL
Z06 Fire and Emergency Service 2000 1500 35.00 2020 2000 0 FULL
Z03 Ward 3 Residential Area 4500 2000 30.00 4891 4500 0 FULL
=====

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Priority Order:
1. Z01 - Municipal Hospital (Priority 40.00)
2. Z04 - Ward 5 Residential Area (Priority 35.00)
3. Z02 - Central Flood Shelter (Priority 35.00)
4. Z05 - School Emergency Shelter (Priority 35.00)
5. Z06 - Fire and Emergency Service (Priority 35.00)
6. Z03 - Ward 3 Residential Area (Priority 30.00)

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Summary
-----
Total Requested Water : 20400.00 L
Total Allocated Water : 21694.00 L
Total Effective Water Delivered: 20400.00 L
Total Unresolved Requirement : 0.00 L
Remaining Water : 306.00 L

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Service Conditions
-----
Full Requirement Satisfied : 6 zone(s)
Minimum Requirement Satisfied: 0 zone(s)
Below Minimum : 0 zone(s)
Completely Unserved : 0 zone(s)

```

Check: total allocated (21694.00 L) does not exceed available supply (22000.00 L). OK.

Zone with the highest unresolved requirement: Z01 (Municipal Hospital) - shortage 0.00 L

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EMERGENCY POTABLE WATER ALLOCATION
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===== CASE 3 =====

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1 #include <stdio.h>
2 #include <string.h>
3
4 #define MAX_ZONES 10
5 #define MAX_NAME 50
6 #define EPSILON 0.000001
7
8
9 typedef struct {
10     char zoneID[10];
11     char service[50];
12     char type[25];
13     double request;
14     double minReq;
15     double lossPer;
16     int prevWat;
17
18     double critical;
19     double priority;
20     int priority;
21
22     double allocated;
23     double effective;
24     double shortage;
25     char condition;
26 } Zone;
27

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25.03_ case1_input.txt all_test_outputs.txt all_test_outputs (1).txt
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===== CASE 3 =====
Insufficient Even For Combined Minimum Requirement
Available Water: 10,000 L

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ALLOCATION RESULT
=====
ID Zone Request Minimum Priority Allocated Delivered Shortage Status
-----
Z01 Municipal Hospital 4000 3000 40.00 3061 3000 1000 MINIMUM
Z04 Ward 5 Residential Area 3800 1800 35.00 2045 1800 2000 MINIMUM
Z02 Central Flood Shelter 3500 2500 35.00 2604 2500 1000 MINIMUM
Z05 School Emergency Shelter 2600 1600 35.00 1684 1600 1000 MINIMUM
Z06 Fire and Emergency Service 2000 1500 35.00 605 599 1401 BELOW MIN
Z03 Ward 3 Residential Area 4500 2000 30.00 0 0 4500 UNSERVED
=====

Priority Order:
1. Z01 - Municipal Hospital (Priority 40.00)
2. Z04 - Ward 5 Residential Area (Priority 35.00)
3. Z02 - Central Flood Shelter (Priority 35.00)
4. Z05 - School Emergency Shelter (Priority 35.00)
5. Z06 - Fire and Emergency Service (Priority 35.00)
6. Z03 - Ward 3 Residential Area (Priority 30.00)

Summary
-----
Total Requested Water : 20400.00 L
Total Allocated Water : 10000.00 L
Total Effective Water Delivered: 9498.89 L
Total Unresolved Requirement : 10901.11 L
Remaining Water : 0.00 L

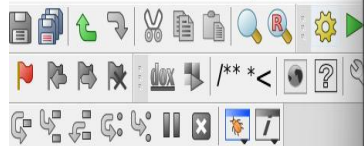
Service Conditions
-----
Full Requirement Satisfied : 0 zone(s)
Minimum Requirement Satisfied: 4 zone(s)
Below Minimum : 1 zone(s)
Completely Unserved : 1 zone(s)

Check: total allocated (10000.00 L) does not exceed available supply (10000.00 L). OK.

Zone with the highest unresolved requirement: Z03 (Ward 3 Residential Area) - shortage 4500.00 L

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EMERGENCY POTABLE WATER ALLOCATION
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===== CASE 4 =====

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```
1 #include <stdio.h>
2 #include <string.h>
3
4 #define MAX_ZONES
5 #define MAX_NAME
6 #define EPSILON
7
8
9 typedef struct {
10     char zoneID[10];
11     char service[20];
12     char type[25];
13     double request;
14     double minReq;
15     double lossPer;
16     int prevWat;
```

25.03_ case1_input.txt all_test_outputs.txt all_test_outputs (1).txt X

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===== CASE 4 =====

Equal Priority Tie-Break Test (Shelter Alpha vs Shelter Beta)

Available Water: 13,500 L

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ALLOCATION RESULT

ID	Zone	Request	Minimum	Priority	Allocated	Delivered	Shortage	Status
ZA	Shelter Alpha	3000	2000	40.00	3158	3000	0	FULL
ZB	Shelter Beta	3000	2000	40.00	3158	3000	0	FULL
Z01	Municipal Hospital	4000	3000	40.00	4082	4000	0	FULL
Z06	Fire and Emergency Service	2000	1500	35.00	2020	2000	0	FULL

Priority Order:

1. ZA - Shelter Alpha (Priority 40.00)
2. ZB - Shelter Beta (Priority 40.00)
3. Z01 - Municipal Hospital (Priority 40.00)
4. Z06 - Fire and Emergency Service (Priority 35.00)

Summary

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-----
Total Requested Water      : 12000.00 L
Total Allocated Water      : 12417.62 L
Total Effective Water Delivered: 12000.00 L
Total Unresolved Requirement : 0.00 L
Remaining Water            : 1082.38 L
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Service Conditions

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-----
Full Requirement Satisfied : 4 zone(s)
Minimum Requirement Satisfied: 0 zone(s)
Below Minimum              : 0 zone(s)
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===== CASE 5 =====
Increased Distribution Loss - Ward 5 Residential Area (Z04) Loss 12% -> 60%
Available Water: 13,500 L

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ALLOCATION RESULT								
ID	Zone	Request	Minimum	Priority	Allocated	Delivered	Shortage	Status
Z01	Municipal Hospital	4000	3000	40.00	3061	3000	1000	MINIMUM
Z04	Ward 5 Residential Area	3800	1800	35.00	4500	1800	2000	MINIMUM
Z02	Central Flood Shelter	3500	2500	35.00	2604	2500	1000	MINIMUM
Z05	School Emergency Shelter	2600	1600	35.00	1684	1600	1000	MINIMUM
Z06	Fire and Emergency Service	2000	1500	35.00	1515	1500	500	MINIMUM
Z03	Ward 3 Residential Area	4500	2000	30.00	135	124	4376	BELOW MIN

```

Priority Order:
1. Z01 - Municipal Hospital (Priority 40.00)
2. Z04 - Ward 5 Residential Area (Priority 35.00)
3. Z02 - Central Flood Shelter (Priority 35.00)
4. Z05 - School Emergency Shelter (Priority 35.00)
5. Z06 - Fire and Emergency Service (Priority 35.00)
6. Z03 - Ward 3 Residential Area (Priority 30.00)

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Summary
Total Requested Water      : 20400.00 L
Total Allocated Water      : 13500.00 L
Total Effective Water Delivered: 10524.43 L
Total Unresolved Requirement : 9875.57 L
Remaining Water            : 0.00 L

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Service Conditions
Full Requirement Satisfied : 0 zone(s)
Minimum Requirement Satisfied: 5 zone(s)
Below Minimum              : 1 zone(s)
Completely Unserved        : 0 zone(s)

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Check: total allocated (13500.00 L) does not exceed available supply (13500.00 L). OK.

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Zone with the highest unresolved requirement: Z03 (Ward 3 Residential Area) - shortage 4375.57 L

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EMERGENCY POTABLE WATER ALLOCATION
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===== CASE 6 =====
Increased Waiting Period - Ward 3 Residential Area (Z03) Waiting 2 -> 8
Available Water: 13,500 L

```



```

1 | #include <stdio.h>
2 | #include <string.h>
3 |
4 | #define MAX_ZONES
5 | #define MAX_NAME
6 | #define EPSILON
7 |
8 |
9 | typedef struct {
10 |     char zoneID[25];
11 |     char service[25];
12 |     char type[25];
13 |     double request;
14 |     double minReq;
15 |     double lossPer;
16 |     int prevWa;
17 |
18 |     double critical;
19 |     double priority;
20 |     int priority;
21 |
22 |     double allocated;
23 |     double effective;
24 |     double shortage;
25 |     char condition;
26 | } Zone;
27 |

```

```

===== CASE 6 =====
Increased Waiting Period - Ward 3 Residential Area (Z03) Waiting 2 -> 8
Available Water: 13,500 L

```

ALLOCATION RESULT								
ID	Zone	Request	Minimum	Priority	Allocated	Delivered	Shortage	Status
Z03	Ward 3 Residential Area	4500	2000	60.00	2590	2383	2117	MINIMUM
Z01	Municipal Hospital	4000	3000	40.00	3061	3000	1000	MINIMUM
Z04	Ward 5 Residential Area	3800	1800	35.00	2045	1800	2000	MINIMUM
Z02	Central Flood Shelter	3500	2500	35.00	2604	2500	1000	MINIMUM
Z05	School Emergency Shelter	2600	1600	35.00	1684	1600	1000	MINIMUM
Z06	Fire and Emergency Service	2000	1500	35.00	1515	1500	500	MINIMUM

```

Priority Order:
1. Z03 - Ward 3 Residential Area (Priority 60.00)
2. Z01 - Municipal Hospital (Priority 40.00)
3. Z04 - Ward 5 Residential Area (Priority 35.00)
4. Z02 - Central Flood Shelter (Priority 35.00)
5. Z05 - School Emergency Shelter (Priority 35.00)
6. Z06 - Fire and Emergency Service (Priority 35.00)

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Summary
-----
Total Requested Water      : 20400.00 L
Total Allocated Water      : 13500.00 L
Total Effective Water Delivered: 12782.61 L
Total Unresolved Requirement : 7617.39 L
Remaining Water            : 0.00 L

```

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Service Conditions
-----
Full Requirement Satisfied : 0 zone(s)
Minimum Requirement Satisfied: 6 zone(s)
Below Minimum              : 0 zone(s)
Completely Unserved        : 0 zone(s)

```

Check: total allocated (13500.00 L) does not exceed available supply (13500.00 L). OK.

Zone with the highest unresolved requirement: Z03 (Ward 3 Residential Area) - shortage 2117.39 L

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===== EMERGENCY POTABLE WATER ALLOCATION =====

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===== CASE 7 (Student-Designed) =====
Available Water Exactly Equals Combined Raw Minimum Requirement (12,400 L)

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